

**Clairity: Smart Restroom Air Quality Monitoring
System Using Gradient Boosting**

USER MANUAL

By Group MEEK

Table of Contents

1. Introduction

- a. What is Clairity?
- b. Key Features

2. Accessing Clairity

- a. Downloading the APK
- b. System Requirements

3. Getting Started

- a. Installing the App (Google Drive Method)
- b. Creating an Account
- c. Logging In

4. Using the Application

- a. Overview
- b. Dashboard Interface
- c. Viewing Real-Time Air Quality Data
- d. Monitoring Pollutants and Alerts
- e. Forecasting Air Quality
- f. Admin Functions

5. Troubleshooting

6. Sample Accounts

Intrroduction

What is CLAIRITY?

Clairity is an AI-powered Android application designed to monitor and predict air quality inside school restrooms. It integrates IoT sensors with machine learning—specifically Gradient Boosting—to detect harmful pollutants, smoke, and poor air quality conditions in real time. Clairity aims to improve safety, hygiene, and awareness in indoor environments through smart monitoring and data-driven forecasting.

Key Features

- Real-Time Air Quality Monitoring: Tracks and displays live air quality data from connected sensors.
- Pollutant Detection: Monitors key gases and particles, including PM_{2.5}, CO, NH₃, H₂S, CH₄, CO₂, and VOCs.
- Fire & Smoke Detection: Identifies cigarette or vape smoke and fire-related air changes.
- Predictive Analysis: Uses Gradient Boosting to forecast air quality trends.
- Admin & User Modes: Admins can manage sensors and users, while general users can view live data and alerts.
- Notifications: Sends alerts for unsafe air conditions or sensor malfunctions.
- Secure Connectivity: Ensures accurate, real-time data exchange between the sensors and the app.

Accessing CLAIRITY

Downloading the APK

The Clairity APK is distributed internally through Google Drive for convenient installation:

1. Open the shared Google Drive link provided by your administrator.
2. Tap the Clairity.apk file.
3. Select Download and wait for the file to finish.

Note: (1). Make sure you have enough storage and a stable internet connection during download. (2) To use the application, use the sample accounts (Refer to the sample account section).

System Requirements

Operating System:

Android 10 or later

Storage: At least 150 MB of free space

Permissions Required:

- Location (for sensor connectivity)
- Notifications (for alerts)- Internet access
- Background activity (for real-time updates)

Getting Started

Installing the App (Google Drive Method)

1. Once the APK is downloaded, locate it in your Downloads folder.
2. Tap the file and select Install.
3. If prompted, enable Install from Unknown Sources.
4. Wait for the installation to complete, then tap Open to launch Clairity.

Creating an Account

1. Launch the Clairity app.
2. Tap Register to create your account.
3. Enter your phone number, email, and other required details.
4. You will receive a One-Time Password (OTP) via SMS or email.
5. Enter the OTP to verify and set your password.

Logging In

1. Open the app and enter your registered phone number or email.
2. Type your password or OTP if applicable.
3. Tap Log In to access your account.

Note: If you are not yet registered, contact the Admin to add your phone number to the database.

Using the Application

Overview

The Clairity interface is designed for simplicity and clarity. The home screen displays current air quality levels, system alerts, and navigation tabs for viewing detailed pollutant data, forecasts, and settings.

Dashboard Interface

Air Quality Index (AQI): Shows overall indoor air quality status (Good, Moderate, Poor). Sensor Readings: Displays live data for each pollutant (PM_{2.5}, CO, NH₃, etc.). Alerts Panel: Highlights fire, smoke, or critical gas detections. Last Updated: Shows the latest data retrieval timestamp.

Viewing Real-Time Air Quality Data

Data updates automatically every few seconds when connected to the sensor network. Users can view pollutant levels and trends in graph form. Color-coded indicators (green, yellow, red) reflect air quality safety levels.

Monitoring Pollutants and Alerts

Tap on any pollutant tile (e.g., CO₂) to view its concentration trends. Alerts are displayed when thresholds are exceeded. Admins receive notifications for severe air quality or potential hazards (e.g., smoke detected).

Forecasting Air Quality

The Forecast section uses historical sensor data and Gradient Boosting predictions. Users can view projected air quality levels for the next several hours. Forecasts help in scheduling maintenance or cleaning routines.

Admin Functions

Admins have exclusive access to:- Adding or removing users- Registering new sensors- Managing database records- Reviewing logs of previous detections and alerts

Note: To access Admin mode, log in with your registered Admin credentials.

Troubleshooting

Issue	Solution
App not installing	Ensure “Install from Unknown Sources” is enabled and sufficient storage space is available.
No sensor data shown	Check sensor connectivity and verify Wi-Fi or Bluetooth connection.
Did not receive OTP	Tap “Resend OTP” or check your internet and SMS service.
App crashes or freezes	Restart your phone and relaunch Clarity. If the issue persists, reinstall the app.

Incorrect air quality readings	Ensure sensors are calibrated and located properly in the restroom.
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Sample Accounts

	Phone Number	Password
Admin Account	9568765481	123456
User Account	9233453243	123456